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Course: DBMS

Normalization Theory — Quick Reference

Normalization is the process of organizing a relational database to reduce data redundancy and improve data integrity. Below are the definitions of all normal forms used in this assignment.

Key Concepts

- Functional Dependency (FD): $X \rightarrow Y$ means attribute(s) X uniquely determines attribute(s) Y
- Candidate Key: A minimal set of attributes that uniquely identifies each tuple
- Prime Attribute: An attribute that is part of ANY candidate key
- Non-Prime Attribute: An attribute that is NOT part of any candidate key
- Superkey: A set of attributes that includes a candidate key

Normal Form	Requirement	Violation Causes
1NF	All attributes are atomic (no repeating groups, no multi-valued attributes)	Multi-valued cells, nested tables
2NF	In 1NF + No partial dependency (non-prime attr must depend on FULL candidate key)	Non-prime attr depends on part of a composite key
3NF	In 2NF + No transitive dependency (non-prime attr depends only on candidate key)	Non-prime attr $A \rightarrow B \rightarrow C$ where B is non-prime
BCNF	For every FD $X \rightarrow Y$, X must be a superkey (stricter than 3NF)	A prime attribute depends on a non-superkey
4NF	In BCNF + No non-trivial multi-valued dependencies	Independent multi-valued facts stored in one table

Question 1: R(ABCDE)

Functional Dependencies: $A \rightarrow B$, $BC \rightarrow D$, $E \rightarrow C$, $D \rightarrow A$

Step 1 — Find All Candidate Keys

We find the closure of each attribute/combination to determine candidate keys.

Attribute Set	Closure	Is Candidate Key?
A	$A^+ = \{A, B\}$	No
E	$E^+ = \{E, C\}$	No
AE	$AE^+ = \{A, E, B, C, D\} = ABCDE$	YES — minimal
BE	$BE^+ = \{B, E, C, D, A\} = ABCDE$	YES — minimal
DE	$DE^+ = \{D, E, A, B, C\} = ABCDE$	YES — minimal

Candidate Keys: {AE}, {BE}, {DE} **Prime Attributes:** A, B, D, E **Non-Prime Attributes:** C

Step 2 — Check 1NF

All attributes are atomic and single-valued. No repeating groups. The relation is already in 1NF.

Status: IN 1NF

Step 3 — Check 2NF

Check for partial dependencies (non-prime attribute depending on a proper subset of a candidate key).

Candidate keys are {AE}, {BE}, {DE}. The only non-prime attribute is C.

- $E \rightarrow C$: E is a proper subset of candidate key {AE}, {BE}, {DE}. This is a PARTIAL DEPENDENCY.

Violation: $E \rightarrow C$ is a partial dependency on C (non-prime) from part of a candidate key. **Anomaly:** Update/Insertion anomaly — C value for a given E is repeated in every tuple that shares that E.

Decompose to achieve 2NF:

- $R1(E, C)$ — carries $E \rightarrow C$
- $R2(A, B, D, E)$ — original minus C

Relation	Attributes	FDs Preserved
R1	E, C	$E \rightarrow C$
R2	A, B, D, E	$A \rightarrow B$, $BC \rightarrow D$ (BC spans both, handle in 3NF), $D \rightarrow A$
Status: IN 2NF		

Step 4 — Check 3NF

Check for transitive dependencies in $R2(A, B, D, E)$.

FDs in $R2$: $A \rightarrow B$, $D \rightarrow A$ (transitive: $D \rightarrow A \rightarrow B$). Since D is prime and A is prime, check non-prime dependencies.

There are no non-prime attributes in $R2$ — all of A, B, D, E are prime (part of candidate keys of $R2$).

For $R1(E, C)$: $E \rightarrow C$, E is the only candidate key, C is non-prime. No transitive dependency.

Status: IN 3NF — no transitive dependencies through non-prime attributes

Step 5 — Check BCNF

For BCNF: every determinant must be a superkey.

$R1(E, C)$: $E \rightarrow C$. E is the candidate key. E is a superkey. BCNF satisfied.

$R2(A, B, D, E)$: FDs: $A \rightarrow B$, $D \rightarrow A$

- $A \rightarrow B$: Is A a superkey of $R2$? A^+ in $R2 = \{A, B\}$. Does not include D or E. A is NOT a superkey.

BCNF Violation in $R2$: $A \rightarrow B$ — A is not a superkey. Anomaly: Redundant storage of B for every tuple with same A value.

Decompose $R2$ for BCNF:

- $R2a(A, B)$ — captures $A \rightarrow B$
- $R2b(A, D, E)$ — remaining, candidate key $\{DE\}$ or $\{AE\}$

Final Relation	Attributes	Candidate Key(s)	BCNF?
R1	E, C	$\{E\}$	YES
R2a	A, B	$\{A\}$	YES — A is superkey
R2b	A, D, E	$\{AE\}, \{DE\}$	YES — $D \rightarrow A$: D subset of candidate key $\{DE\}$, superkey check satisfied

Final BCNF Decomposition: $R1(E,C)$, $R2a(A,B)$, $R2b(A,D,E)$

Question 2: R(ABCDE)

Functional Dependencies: $AB \rightarrow C$, $C \rightarrow D$

Step 1 — Find All Candidate Keys

Attribute Set	Closure	Is Candidate Key?
AB	$AB^+ = \{A, B, C, D\}$	Partial — missing E
ABE	$ABE^+ = \{A, B, C, D, E\} = ABCDE$	YES — minimal

Candidate Key: {ABE} **Prime Attributes:** A, B, E **Non-Prime Attributes:** C, D

Step 2 — Check 1NF

All attributes are atomic. Already in 1NF.

Status: IN 1NF

Step 3 — Check 2NF

Non-prime attributes: C and D. Candidate key is {ABE}. Check if any non-prime attribute depends on a proper subset:

- $AB \rightarrow C$: AB is a proper subset of {ABE}, and C is non-prime. **PARTIAL DEPENDENCY.**
- $C \rightarrow D$: C is not part of the candidate key directly, but C is derived from AB. This will be handled in 3NF.

Violation: $AB \rightarrow C$ is a partial dependency. **Anomaly:** If no tuple exists with ABE, we cannot store the fact that $AB \rightarrow C$.

Decompose to achieve 2NF:

- $R_1(A, B, C)$ — captures $AB \rightarrow C$
- $R_2(A, B, E)$ — remaining (key: ABE)

Relation	Attributes	FDs	Notes
R1	A, B, C	$AB \rightarrow C$	$C \rightarrow D$ still needs checking
R2	A, B, E	None beyond key	Key is {ABE}
Status: IN 2NF			

Step 4 — Check 3NF

In R1(A, B, C): FDs: $AB \rightarrow C$, $C \rightarrow D$. C is non-prime. D is non-prime. $C \rightarrow D$ is a transitive dependency.

Violation in R1: $C \rightarrow D$ is a transitive dependency (C is non-prime, D is non-prime).
Anomaly: Update anomaly — changing D for a given C requires updating all tuples.

Decompose R1 for 3NF:

- R1a(A, B, C) — captures $AB \rightarrow C$, key = {AB}
- R1b(C, D) — captures $C \rightarrow D$, key = {C}

Final Relation	Attributes	Candidate Key	3NF?
R1a	A, B, C	{AB}	YES
R1b	C, D	{C}	YES
R2	A, B, E	{ABE}	YES
Status: IN 3NF			

Step 5 — Check BCNF

For each relation, every FD determinant must be a superkey:

- R1a(A,B,C): $AB \rightarrow C$. AB is the candidate key — superkey. BCNF OK.
- R1b(C,D): $C \rightarrow D$. C is the candidate key — superkey. BCNF OK.
- R2(A,B,E): Key is ABE. No non-trivial FDs beyond the key. BCNF OK.

Final BCNF Decomposition: R1a(A,B,C), R1b(C,D), R2(A,B,E)

Question 3: R(ABCDEF)

Functional Dependencies: $AB \rightarrow C$, $C \rightarrow DE$, $E \rightarrow F$, $F \rightarrow A$

Step 1 — Find All Candidate Keys

Attribute Set	Closure	Is Candidate Key?
AB	$AB^+ = \{A,B,C,D,E,F\} = ABCDEF$	YES — minimal
BC	$BC^+ = \{B,C,D,E,F,A\} = ABCDEF$	YES — minimal
BF	$BF^+ = \{B,F,A,C,D,E\} = ABCDEF$	YES — minimal

Candidate Keys: {AB}, {BC}, {BF} **Prime Attributes:** A, B, C, F **Non-Prime Attributes:** D, E

Step 2 — Check 1NF

All attributes are single-valued and atomic. Already in 1NF.

Status: IN 1NF

Step 3 — Check 2NF

Non-prime attributes: D and E. Candidate keys: {AB}, {BC}, {BF}.

- $AB \rightarrow C \rightarrow DE$: D and E depend on C, not directly on AB. However, $AB \rightarrow DE$ transitively.
- No direct partial dependency: D and E don't depend on proper subsets of any candidate key directly.

There is no partial dependency since D and E are reachable only through full candidate keys (AB or BC or BF).

Status: IN 2NF — no partial dependencies found

Step 4 — Check 3NF

Check transitive dependencies through non-prime attributes:

- $C \rightarrow DE$: C is a prime attribute (part of {BC}). $C \rightarrow D$ and $C \rightarrow E$. D and E are non-prime.
- $E \rightarrow F$: E is non-prime, F is prime. This is allowed in 3NF (the determinant E is checked).

For 3NF violation: a transitive dependency exists where non-prime A depends transitively: $AB \rightarrow C \rightarrow DE$. Since C is prime (part of candidate key {BC}), $C \rightarrow D$ does NOT violate 3NF by the strict definition.

- $F \rightarrow A$: F is prime (part of {BF}), A is prime (part of {AB}). No non-prime involved — no 3NF violation.

Status: IN 3NF — all transitive dependencies involve prime attributes as intermediaries

Step 5 — Check BCNF

For BCNF, every determinant in every FD must be a superkey:

- $AB \rightarrow C$: AB is a candidate key (superkey). OK.
- $C \rightarrow DE$: Is C a superkey? $C^+ = \{C, D, E, F, A\}$ — missing B. C is NOT a superkey. BCNF VIOLATION.
- $E \rightarrow F$: Is E a superkey? $E^+ = \{E, F, A\}$ — not full set. BCNF VIOLATION.
- $F \rightarrow A$: Is F a superkey? $F^+ = \{F, A\}$ — not full set. BCNF VIOLATION.

BCNF Violations: $C \rightarrow DE$ (C not superkey), $E \rightarrow F$ (E not superkey), $F \rightarrow A$ (F not superkey) Anomalies: Redundant storage of D,E whenever the same C appears; update/delete anomalies.

Decompose iteratively for BCNF:

- Split on $C \rightarrow DE$: $R_1(C, D, E)$ and $R_{\text{rest}}(A, B, C, F)$
- In $R_{\text{rest}}(A, B, C, F)$: FDs applicable — $AB \rightarrow C$, $F \rightarrow A$. Check $E \rightarrow F$: E not in R_{rest} , skip.
- $F \rightarrow A$: Is F superkey of R_{rest} ? $F^+ = \{F, A\}$ — not full. Another violation.
- Split R_{rest} on $F \rightarrow A$: $R_2(F, A)$ and $R_3(B, C, F)$
- In $R_1(C, D, E)$: Split on $E \rightarrow F$ not applicable (F not in R_1). $E \rightarrow F$: E is in R_1 . E^+ in $R_1 = \{E\}$. Only FD $E \rightarrow F$ has F outside. Check $C \rightarrow DE$: C is key of R_1 . OK.

Final Relation	Attributes	Candidate Key	FDs	BCNF?
R1	C, D, E	{C}	$C \rightarrow DE$	YES
R2	F, A	{F}	$F \rightarrow A$	YES
R3	B, C, F	{BC}, {BF}	None beyond key	YES

Final Relation	Attributes	Candidate Key	FDs	BCNF?
Final BCNF Decomposition: R1(C,D,E), R2(F,A), R3(B,C,F)				

Question 4: R(ABCD)

Functional Dependencies: $AB \rightarrow CD$, $C \rightarrow A$, $D \rightarrow B$

Step 1 — Find All Candidate Keys

Attribute Set	Closure	Is Candidate Key?
AB	$AB^+ = \{A, B, C, D\} = ABCD$	YES — minimal
CB	$CB^+ = \{C, B, A, D\} = ABCD$	YES — minimal ($C \rightarrow A$, $D \rightarrow B$ via $AB \rightarrow CD$)
AD	$AD^+ = \{A, D, B, C\} = ABCD$	YES — minimal
CD	$CD^+ = \{C, D, A, B\} = ABCD$	YES — minimal

Candidate Keys: {AB}, {BC}, {AD}, {CD} All attributes A, B, C, D are PRIME There are NO non-prime attributes

Step 2 — Check 1NF

All attributes atomic. Already in 1NF.

Status: IN 1NF

Step 3 — Check 2NF

There are no non-prime attributes at all. 2NF trivially satisfied (partial dependencies only apply to non-prime attributes).

Status: IN 2NF — trivially satisfied (no non-prime attributes)

Step 4 — Check 3NF

Similarly, 3NF transitive dependency conditions only apply to non-prime attributes. No non-prime attributes exist.

Status: IN 3NF — trivially satisfied

Step 5 — Check BCNF

For BCNF, every determinant must be a superkey:

- $AB \rightarrow CD$: AB is a candidate key — superkey. OK.
- $C \rightarrow A$: Is C a superkey? $C^+ = \{C, A\}$ — not full set ABCD. NOT a superkey. BCNF VIOLATION.
- $D \rightarrow B$: Is D a superkey? $D^+ = \{D, B\}$ — not full set ABCD. NOT a superkey. BCNF VIOLATION.

BCNF Violations: $C \rightarrow A$ (C not superkey), $D \rightarrow B$ (D not superkey) Note: Despite all attributes being prime and 3NF being satisfied, BCNF is stricter.

Decompose for BCNF (splitting may cause loss of dependency preservation):

- Split on $C \rightarrow A$: $R_1(C, A)$ and $R_2(B, C, D)$
- In $R_2(B, C, D)$: Check $D \rightarrow B$. Is D superkey of R_2 ? $D^+ = \{D, B\}$ in context of R_2 . D is not full key of R_2 . Violation.
- Split R_2 on $D \rightarrow B$: $R_{2a}(D, B)$ and $R_{2b}(C, D)$

Final Relation	Attributes	Candidate Key	FDs	BCNF?
R1	C, A	{C}	$C \rightarrow A$	YES
R2a	D, B	{D}	$D \rightarrow B$	YES
R2b	C, D	{CD}	None beyond key	YES

Note: The FD $AB \rightarrow CD$ cannot be preserved directly in this decomposition. This is acceptable — BCNF does not always guarantee dependency preservation. Lossless join is maintained through the shared keys B and D.

Final BCNF Decomposition: $R_1(C, A)$, $R_{2a}(D, B)$, $R_{2b}(C, D)$

Question 5: R(ABCDE)

Functional Dependencies: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$

Step 1 — Find All Candidate Keys

Note: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ forms a cycle. All of A, B, C, D are mutually determining each other.

Attribute Set	Closure	Is Candidate Key?
A	$A^+ = \{A,B,C,D\}$	Partial — missing E
AE	$AE^+ = \{A,B,C,D,E\} = ABCDE$	YES
BE	$BE^+ = \{B,C,D,A,E\} = ABCDE$	YES
CE	$CE^+ = \{C,D,A,B,E\} = ABCDE$	YES
DE	$DE^+ = \{D,A,B,C,E\} = ABCDE$	YES

Candidate Keys: {AE}, {BE}, {CE}, {DE} **Prime Attributes:** A, B, C, D, E **Non-Prime Attributes:** NONE

Step 2 — Check 1NF

All attributes are atomic. Already in 1NF.

Status: IN 1NF

Step 3 — Check 2NF

No non-prime attributes exist. 2NF trivially satisfied.

Status: IN 2NF — trivially satisfied

Step 4 — Check 3NF

All dependencies: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. For each FD $X \rightarrow Y$, either X is a superkey OR Y is a prime attribute.

- $A \rightarrow B$: A is prime, B is prime. 3NF is satisfied (Y is prime attribute).
- $B \rightarrow C$: B is prime, C is prime. Satisfied.
- $C \rightarrow D$: C is prime, D is prime. Satisfied.
- $D \rightarrow A$: D is prime, A is prime. Satisfied.

Status: IN 3NF

Step 5 — Check BCNF

For BCNF, every determinant must be a superkey:

- $A \rightarrow B$: Is A a superkey? $A^+ = \{A, B, C, D\}$ — missing E . NOT a superkey. BCNF VIOLATION.
- $B \rightarrow C$: Same reasoning, B not a superkey. BCNF VIOLATION.
- $C \rightarrow D$: C not a superkey. BCNF VIOLATION.
- $D \rightarrow A$: D not a superkey. BCNF VIOLATION.

BCNF Violations: All four FDs violate BCNF since none of A , B , C , D alone is a superkey. But note: This is a classic case where 3NF holds but BCNF is violated due to cyclic dependencies.

The cycle $A \leftrightarrow B \leftrightarrow C \leftrightarrow D$ means we cannot decompose losslessly into BCNF without losing some FDs. The standard approach is:

- $R1(A, B)$ — $A \rightarrow B$
- $R2(B, C)$ — $B \rightarrow C$
- $R3(C, D)$ — $C \rightarrow D$ ($D \rightarrow A$ is implied by cycle but not directly preserved)
- $R4(A, E)$ or (D, E) — to include E with some key

Final Relation	Attributes	Candidate Key	BCNF?
R1	A, B	{A}	YES
R2	B, C	{B}	YES
R3	C, D	{C}	YES
R4	A, E	{AE}	YES

Note: $D \rightarrow A$ is losable in this decomposition. BCNF does not always guarantee full dependency preservation. A 3NF decomposition that preserves all FDs would keep $R(A,B,C,D)$ with the original FDs and add $R_5(A,E)$.

BCNF Decomposition: $R_1(A,B)$, $R_2(B,C)$, $R_3(C,D)$, $R_4(A,E)$ 3NF Preserving
Alternative: $R(A,B,C,D) + R_5(A,E)$

Question 6: Normalization of Un-normalized Tables

Each of the four tables below is normalized step-by-step through 1NF, 2NF, 3NF, BCNF, and 4NF where applicable.

Table 1: Event/Organizer Data

Original Un-normalized Table:

EventID	EventName	OrganizerName	OrganizerPhone	Location
E001	Tech Expo 2023	Ayesha Khan	03011223344	Auditorium A
E002	Design Summit	Bilal Tariq	03111223355	Hall B
E003	AI Conference	Saima Riaz	03213456789	Auditorium B
E004	Green Future Forum	Nabeel Shah	03317654321	Seminar Hall
E005	Robotics Challenge	Sana Qureshi	03004561234	Auditorium A
E006	Startup Meetup	Haroon Khalid	03415678900	Conference Room
E007	Women in Tech	Zara Javed	03511223344	Hall C
E008	Blockchain Day	Osama Irfan	03615678987	Hall D
E009	Cloud Computing Forum	Danish Mehmood	03019876543	Auditorium B
E010	Cybersecurity Seminar	Lubna Faisal	03717894563	Hall A

1NF Analysis

All attributes contain atomic values. No repeating groups. One value per cell.

Status: Already in 1NF

Functional Dependencies Identified

- EventID → EventName, OrganizerName, OrganizerPhone, Location
- OrganizerName → OrganizerPhone (each organizer has one phone number)

2NF Analysis

Candidate Key: {EventID}. All non-key attributes fully depend on EventID (single-attribute key, so no partial dependency possible).

Status: Already in 2NF

3NF Analysis

Transitive dependency: EventID $\rightarrow$ OrganizerName $\rightarrow$ OrganizerPhone.

OrganizerPhone depends on OrganizerName (non-prime) which depends on EventID. This is a transitive dependency.

Violation: EventID $\rightarrow$ OrganizerName $\rightarrow$ OrganizerPhone (transitive via non-prime OrganizerName) Anomaly: If an organizer changes phone, every event record for that organizer must be updated.

Decompose for 3NF:

Relation	Attributes	Key	Description
Event	EventID, EventName, OrganizerName, Location	EventID	Event details with organizer reference
Organizer	OrganizerName, OrganizerPhone	OrganizerName	Organizer contact info

Status: IN 3NF

BCNF Analysis

Event table: EventID $\rightarrow$ {EventName, OrganizerName, Location}. EventID is superkey. OK.

Organizer table: OrganizerName $\rightarrow$ OrganizerPhone. OrganizerName is superkey. OK.

Status: IN BCNF

4NF Analysis

No multi-valued dependencies exist in either relation (each event has one organizer, one location).

Status: IN 4NF — no multi-valued dependencies

Final Normalized Relations

EventID	EventName	OrganizerName	Location
E001	Tech Expo 2023	Ayesha Khan	Auditorium A
E002	Design Summit	Bilal Tariq	Hall B
E003	AI Conference	Saima Riaz	Auditorium B
E004	Green Future Forum	Nabeel Shah	Seminar Hall
E005	Robotics Challenge	Sana Qureshi	Auditorium A

(Event table — 10 records total)

OrganizerName	OrganizerPhone
Ayesha Khan	03011223344
Bilal Tariq	03111223355
Saima Riaz	03213456789
Nabeel Shah	03317654321
Sana Qureshi	03004561234
Haroon Khalid	03415678900
Zara Javed	03511223344
Osama Irfan	03615678987
Danish Mehmood	03019876543
Lubna Faisal	03717894563

Table 2: Bookstore / Sales Data

Original Un-normalized Table:

StoreID	StoreName	BookID	BookTitle	QuantitySold
ST01	Reader's Point	BK001	Data Structures	12
ST01	Reader's Point	BK002	Operating Systems	8
ST01	Reader's Point	BK003	Database Systems	10
ST02	Book Haven	BK001	Data Structures	5
ST02	Book Haven	BK004	Python Programming	7
ST03	Scholar's Hub	BK005	Discrete Mathematics	6
ST03	Scholar's Hub	BK006	Software Engineering	9
ST04	Campus Books	BK007	Artificial Intelligence	3
ST04	Campus Books	BK008	Computer Networks	4
ST04	Campus Books	BK009	Machine Learning Basics	2

1NF Analysis

All attributes are atomic. No multi-valued attributes. Already in 1NF.

Status: Already in 1NF

Functional Dependencies Identified

- StoreID → StoreName (each store has one name)
- BookID → BookTitle (each book has one title)
- {StoreID, BookID} → QuantitySold (sales quantity is per store per book)

2NF Analysis

Candidate Key: {StoreID, BookID}. Check partial dependencies on non-prime attributes:

- StoreID → StoreName: StoreName depends only on StoreID (part of key). PARTIAL DEPENDENCY.
- BookID → BookTitle: BookTitle depends only on BookID (part of key). PARTIAL DEPENDENCY.

Violations: $\text{StoreID} \rightarrow \text{StoreName}$ and $\text{BookID} \rightarrow \text{BookTitle}$ are both partial dependencies. **Anomalies:** StoreName is repeated for every book sold at that store. BookTitle is repeated for every store selling that book.

Decompose for 2NF:

Relation	Attributes	Key
Store	StoreID, StoreName	StoreID
Book	BookID, BookTitle	BookID
Sales	StoreID, BookID, QuantitySold	{StoreID, BookID}

Status: IN 2NF

3NF & BCNF Analysis

Store(StoreID, StoreName): $\text{StoreID} \rightarrow \text{StoreName}$. Single FD with superkey. 3NF and BCNF satisfied.

Book(BookID, BookTitle): $\text{BookID} \rightarrow \text{BookTitle}$. Single FD with superkey. 3NF and BCNF satisfied.

Sales(StoreID, BookID, QuantitySold): Composite key. QuantitySold depends on full key. No transitive dependencies.

Status: IN 3NF and BCNF

4NF Analysis

In Sales table: A store sells many books, a book is sold at many stores. These are independent facts. If modeled as multi-valued dependencies ($\text{StoreID} \twoheadrightarrow \text{BookID}$ and $\text{StoreID} \twoheadrightarrow \text{QuantitySold}$), but QuantitySold is uniquely determined by {StoreID, BookID}, so no non-trivial multi-valued dependency exists.

Status: IN 4NF

Final Normalized Relations

StoreID	StoreName
ST01	Reader's Point

StoreID	StoreName
ST02	Book Haven
ST03	Scholar's Hub
ST04	Campus Books

BookID	BookTitle
BK001	Data Structures
BK002	Operating Systems
BK003	Database Systems
BK004	Python Programming
BK005	Discrete Mathematics

StoreID	BookID	QuantitySold
ST01	BK001	12
ST01	BK002	8
ST01	BK003	10
ST02	BK001	5
ST02	BK004	7

Table 3: Vehicle / Usage Log

Original Un-normalized Table:

VehicleID	VehicleType	EmployeeID	EmployeeName	DateUsed
V01	Sedan	EMP001	Umar Farooq	2023-10-10

VehicleID	VehicleType	EmployeeID	EmployeeName	DateUsed
V01	Sedan	EMP002	Samina Nisar	2023-10-12
V02	Van	EMP001	Umar Farooq	2023-10-15
V03	SUV	EMP003	Zeeshan Khalid	2023-10-18

1NF Analysis

All attributes atomic. No repeating groups. Already in 1NF.

Status: Already in 1NF

Functional Dependencies Identified

- VehicleID → VehicleType (vehicle type determined by vehicle ID)
- EmployeeID → EmployeeName (employee name determined by employee ID)
- {VehicleID, EmployeeID, DateUsed} → (entire record) (composite key for usage log)

2NF Analysis

Candidate Key: {VehicleID, EmployeeID, DateUsed}. Check partial dependencies:

- VehicleID → VehicleType: Depends only on VehicleID. PARTIAL DEPENDENCY.
- EmployeeID → EmployeeName: Depends only on EmployeeID. PARTIAL DEPENDENCY.

Violations: VehicleType and EmployeeName partially depend on subsets of the composite key. Anomalies: VehicleType repeated for every usage of the same vehicle. EmployeeName repeated for every vehicle use by same employee.

Decompose for 2NF:

Relation	Attributes	Key
Vehicle	VehicleID, VehicleType	VehicleID
Employee	EmployeeID, EmployeeName	EmployeeID
UsageLog	VehicleID, EmployeeID, DateUsed	{VehicleID, EmployeeID, DateUsed}

Status: IN 2NF

3NF & BCNF Analysis

All three relations have simple key structures. No transitive dependencies exist. Each determinant is a superkey.

Status: IN 3NF and BCNF

4NF Analysis

UsageLog(VehicleID, EmployeeID, DateUsed): A vehicle can be used by multiple employees on different dates, and an employee can use multiple vehicles. If VehicleID $\rightarrow\rightarrow$ EmployeeID and VehicleID $\rightarrow\rightarrow$ DateUsed independently existed, that would be a 4NF violation. However, DateUsed is tied to a specific (vehicle, employee) pair — it's not independent. No non-trivial MVD.

Status: IN 4NF

Final Normalized Relations

VehicleID	VehicleType
V01	Sedan
V02	Van
V03	SUV

EmployeeID	EmployeeName
EMP001	Umar Farooq
EMP002	Samina Nisar
EMP003	Zeeshan Khalid

VehicleID	EmployeeID	DateUsed
V01	EMP001	2023-10-10
V01	EMP002	2023-10-12

VehicleID	EmployeeID	DateUsed
V02	EMP001	2023-10-15
V03	EMP003	2023-10-18

Table 4: Supplier / Inventory Data

Original Un-normalized Table (16 records):

SupplierID	SupplierName	ItemID	ItemName	DeliveryDate
SUP01	Galaxy Traders	IT001	LED Monitors	2023-09-01
SUP01	Galaxy Traders	IT002	Wireless Mouse	2023-09-03
SUP01	Galaxy Traders	IT003	Keyboards	2023-09-04
SUP02	Tech Suppliers	IT004	Laptop Chargers	2023-09-05
SUP02	Tech Suppliers	IT005	HDMI Cables	2023-09-06
SUP02	Tech Suppliers	IT006	Power Strips	2023-09-07
SUP03	Office Essentials	IT007	Whiteboards	2023-09-08
SUP03	Office Essentials	IT008	Markers Pack	2023-09-09
SUP03	Office Essentials	IT009	Sticky Notes	2023-09-10
SUP04	SmartTech Deals	IT010	Laptop Bags	2023-09-11
SUP04	SmartTech Deals	IT011	External Hard Drives	2023-09-12
SUP04	SmartTech Deals	IT012	USB Hubs	2023-09-13
SUP05	PC World	IT013	Monitors (Curved)	2023-09-14
SUP05	PC World	IT014	Graphic Tablets	2023-09-15
SUP05	PC World	IT015	Docking Stations	2023-09-16
SUP05	PC World	IT016	Webcam HD	2023-09-17

1NF Analysis

All attributes are atomic and single-valued. No multi-valued attributes or nested tables.

Status: Already in 1NF

Functional Dependencies Identified

- $\text{SupplierID} \rightarrow \text{SupplierName}$ (each supplier has one name)
- $\text{ItemID} \rightarrow \text{ItemName}$ (each item has one name)
- $\{\text{SupplierID}, \text{ItemID}\} \rightarrow \text{DeliveryDate}$ (a specific supplier delivers a specific item on a date)

2NF Analysis

Candidate Key: $\{\text{SupplierID}, \text{ItemID}\}$. Check partial dependencies:

- $\text{SupplierID} \rightarrow \text{SupplierName}$: Partial — SupplierName depends on only part of composite key.
- $\text{ItemID} \rightarrow \text{ItemName}$: Partial — ItemName depends on only part of composite key.

Violations: SupplierName and ItemName are partially dependent on subsets of the composite key. Anomalies: SupplierName repeated for each item supplied. Deleting all items of a supplier loses supplier name.

Decompose for 2NF:

Relation	Attributes	Key
Supplier	SupplierID, SupplierName	SupplierID
Item	ItemID, ItemName	ItemID
Delivery	SupplierID, ItemID, DeliveryDate	$\{\text{SupplierID}, \text{ItemID}\}$

Status: IN 2NF

3NF & BCNF Analysis

Supplier(SupplierID, SupplierName): Single FD with superkey. 3NF and BCNF satisfied.

Item(ItemID, ItemName): Single FD with superkey. 3NF and BCNF satisfied.

Delivery(SupplierID, ItemID, DeliveryDate): No non-trivial FDs beyond the composite key. 3NF and BCNF satisfied.

Status: IN 3NF and BCNF

4NF Analysis

Delivery table: SupplierID →→ ItemID (a supplier supplies many items) and SupplierID →→ DeliveryDate — but DeliveryDate is specific to (SupplierID, ItemID), not independent of ItemID. If a supplier could supply any item on any date independently, that would be a 4NF violation. However here, each delivery record is a unique (Supplier, Item, Date) triple with no independent MVDs.

Status: IN 4NF — no non-trivial multi-valued dependencies

Final Normalized Relations

SupplierID	SupplierName
SUP01	Galaxy Traders
SUP02	Tech Suppliers
SUP03	Office Essentials
SUP04	SmartTech Deals
SUP05	PC World

ItemID	ItemName
IT001	LED Monitors
IT002	Wireless Mouse
IT003	Keyboards
IT004	Laptop Chargers
IT005	HDMI Cables
IT006	Power Strips
IT007	Whiteboards
IT008	Markers Pack

SupplierID	ItemID	DeliveryDate
SUP01	IT001	2023-09-01
SUP01	IT002	2023-09-03
SUP01	IT003	2023-09-04
SUP02	IT004	2023-09-05
SUP02	IT005	2023-09-06
SUP02	IT006	2023-09-07

(Delivery table — 16 records total)

Summary: Normalization Results

Question	Relation(s)	Original NF	Final NF	Key Observations
Q1	R(ABCDE)	1NF	BCNF	Partial dep $E \rightarrow C$; BCNF split into R1(E,C), R2a(A,B), R2b(A,D,E)
Q2	R(ABCDE)	1NF	BCNF	Partial dep $AB \rightarrow C$; transitive $C \rightarrow D$; split into R1a(A,B,C), R1b(C,D), R2(A,B,E)
Q3	R(ABCDEF)	1NF	BCNF	$C \rightarrow DE$, $E \rightarrow F$, $F \rightarrow A$ violate BCNF; split into R1(C,D,E), R2(F,A), R3(B,C,F)
Q4	R(ABCD)	3NF	BCNF	All prime attributes; 3NF trivially satisfied; BCNF split into R1(C,A), R2a(D,B), R2b(C,D)
Q5	R(ABCDE)	3NF	BCNF	Cyclic FDs; 3NF satisfied; BCNF loses $D \rightarrow A$; alt: keep R(A,B,C,D)+R5(A,E) for 3NF
Q6-T1	Event/Organizer	1NF	4NF	3NF violation: OrganizerPhone transitive; split Event + Organizer tables
Q6-T2	Bookstore/Sales	1NF	4NF	2NF violations: StoreName, BookTitle partial deps; split Store + Book + Sales
Q6-T3	Vehicle/Usage	1NF	4NF	2NF violations: VehicleType, EmployeeName partial deps; split into 3 tables
Q6-T4	Supplier/Inventory	1NF	4NF	2NF violations: SupplierName, ItemName partial deps; split into 3 tables